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[SMART SOIL IRRIGATION SYSTEM]

Revision History

Date	Description	Author	Comments
27-04-2026	<Version 1>	Ammara Mukhtar	<First Revision>

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	MR. Mustafa Hameed	Supervisor, DIT	27-04-2026

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1. Introduction

Agriculture is one of the most important sectors in the world, and water is one of the most valuable natural resources for plant growth. Traditional irrigation methods depend heavily on farmer experience and manual effort, which may result in over-irrigation, under-irrigation, and unnecessary wastage of water. To overcome these challenges, the Smart Soil Irrigation System introduces automation and intelligence into irrigation practices. This system ensures that plants receive the required amount of water based on actual soil moisture conditions rather than estimation.

The Smart Soil Irrigation System uses a soil moisture sensor to continuously monitor the water content in soil. When the soil becomes dry, the sensor sends updated readings to the microcontroller. The microcontroller processes the data and automatically controls the water pump. Once the soil reaches a sufficient moisture level, the pump is turned OFF. Thus, the system provides accurate, real-time, condition-based irrigation to prevent both water deficiency and wastage.

To make the system more practical and advanced, wireless communication is integrated using a Wi-Fi module. This allows farmers to monitor the moisture status, pump condition, and system mode through a mobile or web interface, even when they are not physically present. Manual override control is also available, enabling users to turn ON/OFF the pump remotely with a single click.

This system runs on a main power supply or a battery. For long-term deployment in large farms or remote areas, the system can be integrated with solar panels, ensuring sustainable energy usage. The microcontroller is programmed using C/C++, which provides fast execution and compatibility with embedded hardware. This irrigation automation solution not only reduces human effort but also improves crop yield, saves water, reduces electricity consumption, and supports farmers in making their fields smarter and more productive.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to clearly define the complete set of **functional**, **non-functional**, **hardware**, and **software** requirements for the Smart Soil Irrigation System. The goals of this document are:

- To establish a mutual understanding between developers, supervisors, and stakeholders.
- To provide a solid foundation for the design, development, testing, deployment, and evaluation of the system.
- **To ensure that the system delivers:**
    - Water efficiency through automatic irrigation control
    - Higher agricultural productivity
    - Remote monitoring and ease of use for farmers
    - Reduced human involvement for improved reliability

- This document ensures clarity, correctness, and completeness before starting implementation.

1.2 Scope

This system enables continuous monitoring and intelligent watering in multiple real-world environments such as:

- Large agricultural farms
- Greenhouses and nurseries
- Home gardens and rooftop plants
- Parks and landscaped areas

The key capabilities of this system include:

- Automatic irrigation based on moisture levels
- Remote monitoring and control via the internet
- Live display of soil moisture percentage
- Pump ON/OFF automation with safety limits
- Battery or solar-powered operation
- Data logging for decision making
- User authentication for secure access

Benefits to users:

Benefit	Description
Water Conservation	**Eliminates unnecessary watering**
Higher Yield	**Maintains optimal soil conditions**
Low Labor	**Reduces manual workload for farmers**
Energy Savings	**Solar implementation reduces electricity usage**
Accessible Monitoring	**Farmers can check conditions anytime from anywhere**

1.3 Definitions, Acronyms, and Abbreviations.

Term	Description
1. Smart Irrigation	Automatic supply of water based on real-time soil needs
2. Sensor Calibration	Adjustment of sensor output to match actual soil conditions
3. IoT	Internet of Things — devices connected to cloud for communication
4. Relay Driver	Switching circuit used to operate AC water pump
5. Threshold	Minimum required soil moisture for plant needs
6. Auto Mode	System decides irrigation based on sensor readings
7. Manual Mode	User fully controls irrigation from remote application
8. ESP8266/ESP32	Wi-Fi supported microcontroller boards
9. Firmware	Embedded software running on microcontroller
10. Cloud Server	Remote platform for data storage and access

1.4 References

These sources support the technical and theoretical background of the project:

- Arduino Microcontroller Documentation — official programming and pin-interface guidelines.
- Soil Moisture Sensor Datasheet — electrical characteristics and calibration methods.
- IoT-based Smart Irrigation Techniques — research publications and case studies.

- Wi-Fi Communication Protocol Standards — IEEE wireless network usage guidelines.
- Tutorials on relay interfacing with microcontrollers — electronic switching methodology.
- Solar energy system integration research for agricultural automation.

1.5 Overview

This SRS document is organized into structured sections:

- **Section 1:** Introduces the purpose, scope, and terminology of the system.
- **Section 2:** Describes the overall system perspective, product functions, constraints, and assumptions.
- **Section 3:** Contains detailed descriptions of functional requirements, hardware/software interfaces, use cases, performance parameters, and database needs.
- Additional diagrams (use-case, DFD, architecture, flowcharts) support technical design clarity.

This document acts as a blueprint to ensure the development of a reliable, efficient, secure, and user-friendly Smart Soil Irrigation System.

2. The Overall Description

The Smart Soil Irrigation System is an IoT-based embedded system designed to automate agricultural irrigation and provide remote monitoring capabilities. The system continuously senses the soil moisture level and operates a water pump based on real-time soil conditions. Unlike traditional irrigation where farmers must manually monitor and water crops, this system makes irrigation intelligent and automatic. The main components of the system include a soil moisture sensor, a microcontroller (Arduino with Wi-Fi module such as ESP8266), relay module, water pump, and optional solar power supply. The system uploads moisture values to a cloud platform which allows users to control irrigation from a mobile or web interface. This system is reliable, energy-efficient, and highly beneficial for farms located in remote areas.

2.1 Product Perspective

The Smart Soil Irrigation System is a **stand-alone automation product** that can also be expanded into larger IoT-based smart agriculture ecosystems. It is designed as a replacement and improvement over manual irrigation systems. The system uses data from the real environment (soil moisture sensor) and automatically controls the irrigation pump through programmed decision-making. By integrating Wi-Fi communication, the system becomes more interactive, allowing farmers to receive real-time information from anywhere.

This system includes the following modules:

Module	Description
Sensing Module	Soil Moisture Sensor monitors soil condition
Control Module	Microcontroller processes sensor input and triggers actions
Power Module	Battery or solar panel to power electronics and pump
Communication Module	Wi-Fi for cloud connectivity and remote monitoring
Actuation Module	Relay + Pump to control irrigation automatically

The product can be installed on small garden beds or extended across wide agricultural fields with additional sensors and pumps.

2.1.1 Operations

- The system performs its operations continuously through the following process:
- **Soil Monitoring Phase**
  - Soil moisture sensor collects humidity level every few seconds
  - The microcontroller interprets data and checks threshold values
- **Decision-Making Phase**
  - If soil moisture < threshold → Pump ON
  - If soil moisture ≥ required level → Pump OFF
  - Safety timer prevents long or unnecessary pump running
- **Remote Monitoring Phase**
  - Moisture level is uploaded to cloud
  - Farmer views data on mobile/web dashboard
  - Manual commands can override automatic mode
- **Power Management Phase**
  - System runs on DC supply or solar power
  - Low-power mode used during inactivity

- This automated cycle ensures irrigation only when needed, improving water efficiency.

2.1.2 Site Adaptation Requirements

The system must be installed and adapted according to real agricultural or gardening conditions:

Requirement Category	Details
Environmental	**Electronics must be protected from heat, dust, insects, and rain (IP-rated waterproof casing recommended)**
Placement of Sensor	**Sensor must be inserted 5–10 cm deep in soil for accurate readings**
Pump Installation	**Pump installed near water source with proper hose connections**
Network Requirements	**Wi-Fi coverage is required for cloud features; alternative GSM module can be used in remote areas**
Power Requirements	**Solar panels or rechargeable batteries preferred in rural areas to maintain continuous operation**

Large farms may require multiple sensors for accurate moisture distribution feedback.

2.2 Product Functions

This system provides the following main functions:

Feature Category	Detailed Functionality	Benefit
Automatic Irrigation	Pump activation/deactivation based on soil dryness	Saves water & eliminates manual effort
Remote Monitoring	View live moisture percentage on mobile/web app	Farmer can monitor status anytime
Remote Manual Control	Command pump ON/OFF remotely	Full flexibility & convenience

Feature Category	Detailed Functionality	Benefit
Alert Notifications	Low moisture alerts sent to user	Prevents crop stress
Data Logging	Keeps history of moisture & pump events	Helps in farming decisions
Self-Protection	Pump timeout & fault detection	Ensures system safety
Solar Integration	Works independently off-grid	Useful in rural regions

These functions ensure optimal plant growth and resource management.

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2.3 User Characteristics

User Type	Technical Knowledge	Responsibilities	Example User
Primary User	**Low — Normal farmers / Gardeners**	**Checking moisture status, controlling pump**	**Farmer, Gardener**
Secondary User	**Medium — Basic electronics skill**	**System installation & maintenance**	**Technician**
Technical/Developer User	**High — Programming + Networking**	**Updating firmware, cloud settings**	**Engineer**

2.4 GeneralConstraints

Constraint Category	Impact on System
Sensor Accuracy Variation	Different soil types (clay, sand) affect readings → requires calibration
Climate & Weather Exposure	Electronics must survive heat, moisture, insects, dust

Constraint Category	Impact on System
Power Supply Limitations	Pump requires high current → battery capacity must be large
Internet Connectivity	No Wi-Fi → Remote monitoring disabled, only Auto Mode active
Relay Load Capacity	Pump motor should not exceed rated current load
Component Lifespan	Harsh environments reduce durability of sensors and pump

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2.5 Assumptions and Dependencies

Project Assumptions:

- Moisture sensor is correctly calibrated with the crop's soil type
- Farmer/family has a smartphone to access control app
- Wi-Fi network or alternative connectivity is available
- Water source (tank/pipe) is functioning and always filled
- System installed correctly according to provided guidelines

External Dependencies:

- Quality and lifetime of external components (pump, pipes, sensors)
- Cloud server uptime and security policies
- Availability of sunlight for solar panel-based systems
- Microcontroller firmware stability
- If any dependency fails, irrigation performance reduces.

3. Specific Requirements

This section defines the complete system behavior for the Smart Soil Irrigation System. It includes interaction requirements, functional automation rules, user interaction, safety features, network behavior, and quality expectations for a fully deployable IoT agricultural solution.

The requirements are derived from:

- User needs

- Agricultural irrigation domain standards
- Environmental constraints
- Embedded system reliability concerns
- IoT communication behavior
- The SMART requirements below must be Specific, Measurable, Achievable, Relevant, and Testable.

3.1 External Interface Requirements

3.1.1 System Interfaces

The Smart Irrigation System must interact with the following subsystems:

Subsystem	Interaction Role	Dependency	Failure Handling
Soil Moisture Sensor	Measure soil humidity levels	Sensor integrity	Switch to default irrigation cycle if failure detected
Relay + Pump Unit	Performs irrigation	Power stability	Immediate shut-off on overload
Cloud Server	IoT communication	Internet connectivity	Auto local fallback logic
Mobile/Web App	Remote control & monitoring	Cloud success	Manual control disabled offline

Detailed Designs:

- Moisture sensor communicates through ADC to MCU using voltage output values converted to moisture percentage.
- Relay communicates using digital logic signals; pump uses inductive load, so protection diode is required.
- Cloud communication uses MQTT, optimized for machine-to-machine data.
- Why: To ensure safe, accurate, continuous communication between sensing → intelligence → action → monitoring flow.

3.1.2 Interfaces

The user interface must be designed for farmer-friendly usage:

UI Functional Requirements

- Live moisture display: “% Moisture”, “Soil Status: Dry / Optimal / Wet”
- Pump ON/OFF control button visible at all times
- **Mode selects:** Automatic / Manual
- Moisture threshold slider/input box (0–100%)
- Historical charts: hourly, daily, weekly trends
- Push notifications when system detects dryness for >180 seconds
- User profile: name, farm details, system ID
- Device health display: Wi-Fi strength, battery percentage (if solar)

UI Usability Requirements

- Click actions must not exceed 2 steps
- Font must be readable in sunlight
- Icons instead of text for low-literacy users
- Multilingual support (e.g., English, Urdu)

UI Safety Requirements

- Critical functions require confirmation pop-ups:
  - “Start Pump?” → YES/NO
- Auto logout after 10 min of inactivity

UI Accessibility

- Mobile responsive layout including low-brightness mode for night workers

  Why:
  Farmers often work in harsh outdoor conditions → UI must remain usable anywhere.

3.1.3 Hardware Interfaces

Component	Pin Type & Role	Max Ranges	Environmental Requirement	Justification
Moisture Sensor	Analog input	0–5V	Waterproof coating	Soil corrosion risk

Component	Pin Type & Role	Max Ranges	Environmental Requirement	Justification
Microcontroller	Wi-Fi + GPIO	3.3V logic	Heat resistant enclosure	Protect from UV
Relay Module	Switch actuator	10A max	Dry location	Relay sparks risk
Water Pump	High-power actuator	AC/DC 220V/12V	Keep near water	Pump access convenience
Solar Panel (Optional)	DC Power Source	12–18V	Direct sunlight exposure	Off-grid operation

Electrical Safety Requirements:

- Fuse in line with pump
- Reverse polarity protection
- Grounding required for AC systems
- Why:Prevent electrical fire hazards and ensure farm safety.

3.1.4 Software Interfaces

Software Layer	Task	Technology
Firmware	Sensor reading, automation logic	Arduino C/C++
IoT Protocol	Message publishing/subscribing	MQTT
Cloud Platform	Data processing and storage	Firebase/Blynk
UI Dashboard	User interaction	HTML/JS or Mobile App
Data Visualization	Graphs, charts	Time-series plotting API

Constraints:

- Firmware must be modular for future upgrade (OTA update optional)
- Data packets compressed for low bandwidth
- **Why:**Ensures **scalability** and **long-term maintainability**

3.1.5 Communications Interfaces

Feature	**Requirement**
Standard	IEEE 802.11 b/g/n Wi-Fi
Data Rate	512 kbps optimized for IoT
Encryption	TLS/SSL recommended
Reconnect Logic	Auto retry every 10 seconds
Offline Mode	System continues automatic irrigation

.

3.2 Functional Requirements

Functional requirements ensure precise behavior in all scenarios.

🌱 3.2.1 Feature #1: Soil Moisture Monitoring

ID	**Detailed Requirement**	**Measurement Criteria**	**Reason**
FR-01	Sample soil moisture every 5 seconds	Timer accuracy ± 1s	Real-time precision
FR-02	Convert raw ADC → % moisture	Error ≤ 5%	Uniform data
FR-03	Identify soil zone as Dry/Normal/Wet	Based on threshold logic	Clear decisions
FR-04	Detect faulty sensor reading	If constant 0% or 100% >1 min	Avoid false watering
FR-05	Upload data to cloud continuously	No more than 3-second delay	Remote monitoring
FR-06	Store minimum 7 days log	At least 1000 data points	Analysis support

💧 3.2.2 Feature #2: Automatic Irrigation Control

ID	**Requirement**	**Safety Mechanism**	**Why**
FR-07	Pump must ON when soil dry	Hysteresis to avoid rapid	Prevent motor

ID	Requirement	Safety Mechanism	Why
		cycling	burnout
FR-08	Pump must OFF when targeted % achieved	Verified double reading	Avoid root rot
FR-09	Maintain continuous pumping ≤10 minutes	Timeout auto-stop	Prevent flooding
FR-10	Log each irrigation event	Store timestamp + duration	Helps in crop studies
FR-11	Lockout if sensor fault	Pump OFF	Safety priority

3.2.3 Feature #3: Remote Manual Control

ID	Requirement	Success Condition
FR-12	Authorized user can manually turn pump ON/OFF	Cloud command success
FR-13	Pump status feedback to user	Remote UI updates within 3s
FR-14	Auto mode temporarily paused	Resume controls after 10 min without input

3.3 Use Cases

3.3.1 Use Case 1 – Automatic Irrigation

Element	Description
Primary Actor	Environment + System
Goal	Maintain optimal crop hydration
Preconditions	Auto mode ON, sensor healthy
Main Flow	Soil dry → Pump ON → Soil normal → Pump OFF
Alternate Flow	Network down → Continue local processing
Postcondition	Water savings + healthy soil moisture

Element	Description
Exceptions	System enters safe mode if pump stuck ON

3.3.2 Use Case 2 – Manual Override Control

Element	Description
Actor	User (Farmer)
Purpose	Full control of irrigation
Trigger	User presses ON/OFF
Precondition	User authenticated
Error Handling	Timeout + confirmation alerts
Security	Display intrusion log for unauthorized attempts

3.4 Classes / Objects (Object-Oriented Modeling)

3.4.1 Sensor Class

- Attributes: moistureValue, sensorStatus, calibrationFactor
- Methods:
  - readMoisture()
  - calibrate()
  - detectError()

3.4.2 Controller Class

- Attributes: pumpState, mode, thresholdValue
- Methods:
  - checkMoisture()
  - applyDecision()
  - sendToCloud()
  - offlineFallback()

...

3.5 Non-Functional Requirements

3.5.1 Performance Requirements

- *Response time for pump command:* ***≤ 2 sec***
- *Data loss tolerance:* ***≤ 5%***
- *System must support* ***>100 users*** *(multi-greenhouse expansion)*

3.5.2 Reliability Requirements

- *Uptime:* ***≥ 98% over 30 days***
- *Watchdog timer resets MCU on software failure*

3.5.3 Availability Requirements

- *Power backup ensures* ***24/7*** *operation*
- *Offline Auto Mode for* ***100% irrigation continuity***

3.5.4 Security Requirements

- *Password hashing & encryption on communication*
- *Role-based access control*
- *Intrusion detection logs*

3.5.5 Maintainability

- *Sensors replaceable without firmware change*
- *Modular code → future AI upgrades possible*

3.5.6 Portability

- *System relocatable in < 20 minutes*
- *Supports any soil type after calibration*

3.6 Inverse Requirements

None

3.7Logical Database Requirements

Entities involved:

- **User Table**
- **MoistureLog Table**
- **PumpControlLog Table**

Data Volume Capacity:

- **Designed for 100,000 entries minimum**

Backup Strategy:

- **Cloud auto-backup daily**
- **Exportable in CSV for research reports**

structure.

3.8Design Constraints

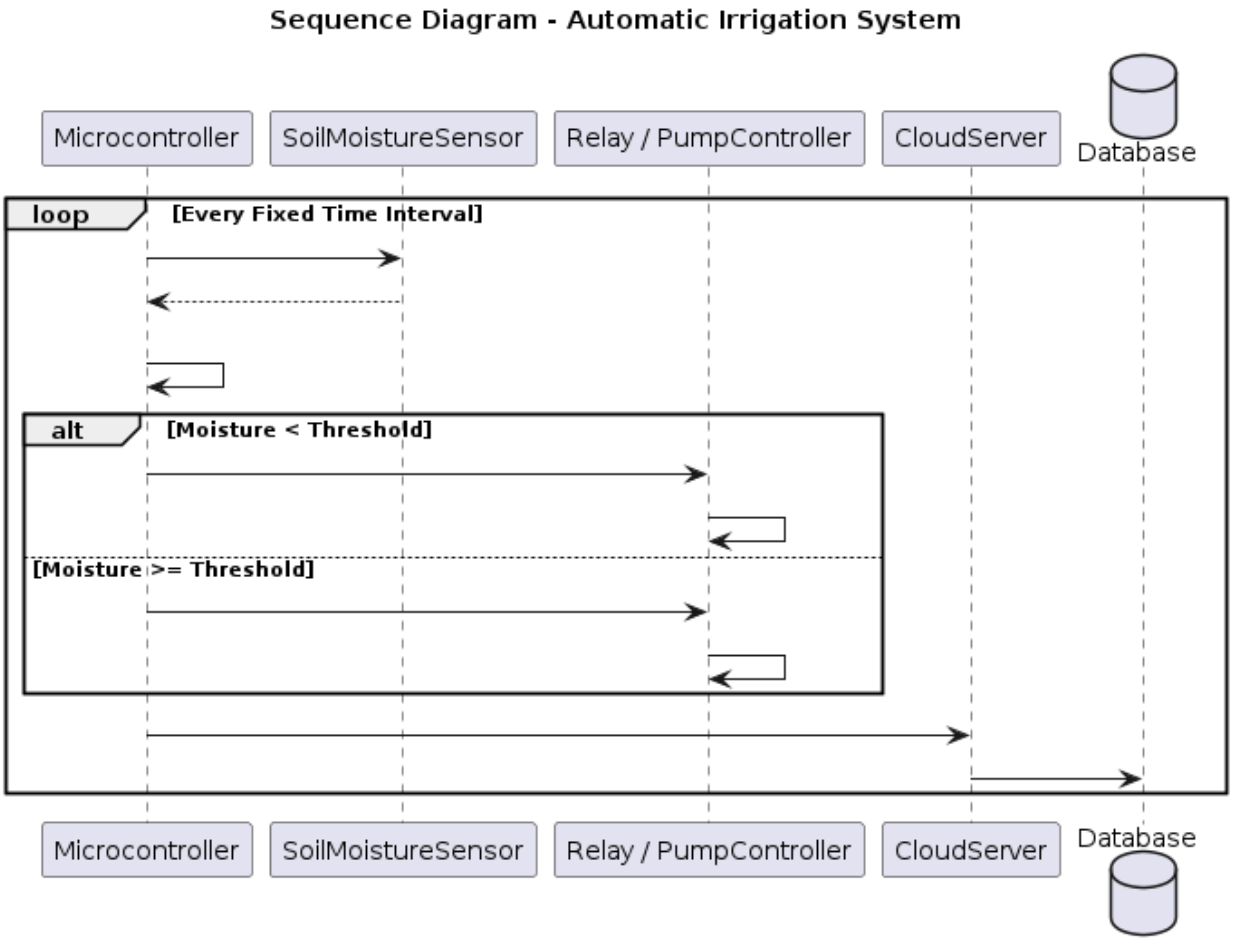
Constraint	Impact
Pump specs must match farm irrigation demand	Hardware cost varies with farm size
Wi-Fi limited coverage	Possible use of long-range LoRaWAN
Harsh UV and dust outdoors	Enclosure must be IP65
Limited solar energy in winter	Battery sizing must compensate

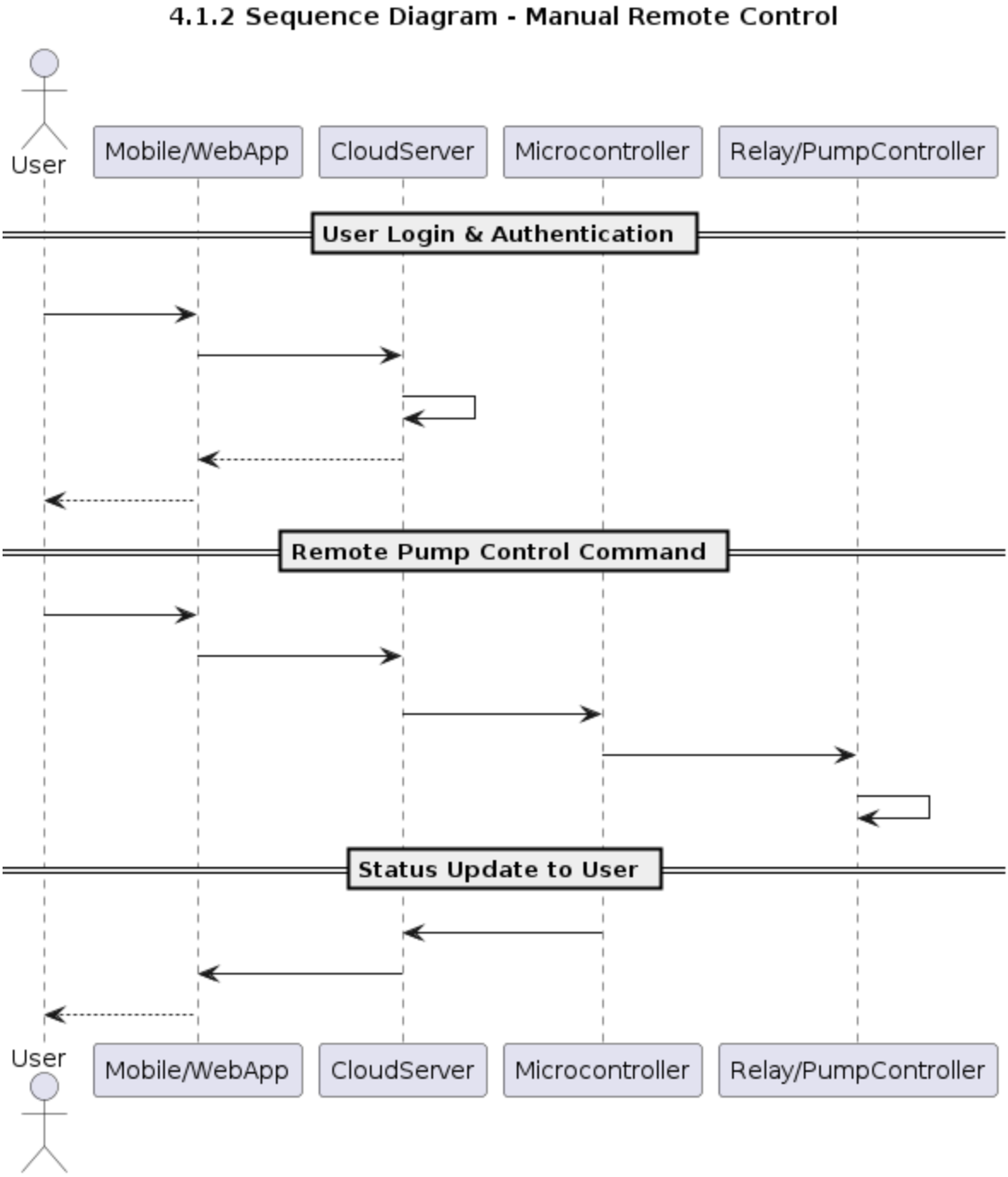
Industry safety practices must be followed due to water + electricity combination hazards.

4. Analysis Models

List all analysis models used in developing specific requirements previously given in this SRS. Each model should include an introduction and a narrative description. Furthermore, each model should be traceable the SRS's requirements.

4.1 Sequence Diagrams

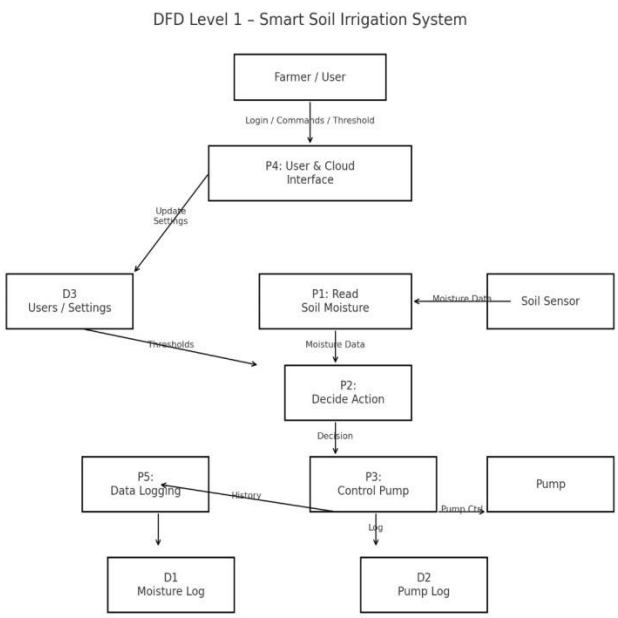


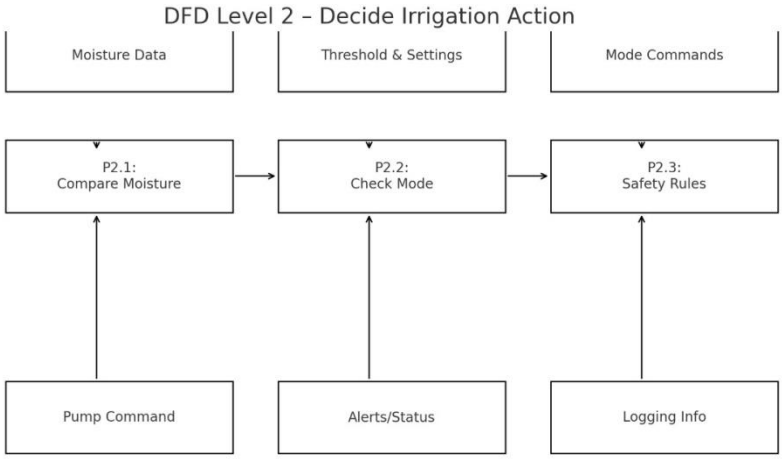


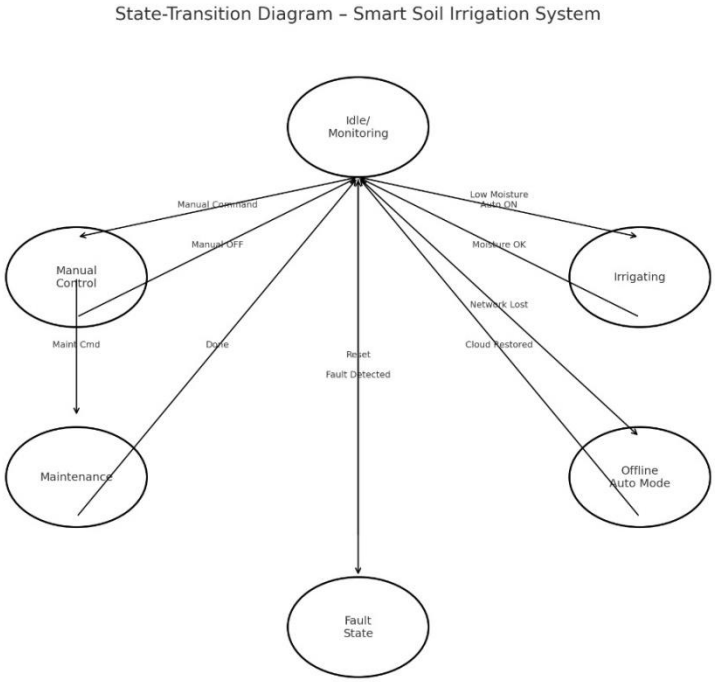
4.2 Data Flow Diagrams (DFD)

DFD(Level 0)

```
        +-------------------------+
        |       Farmer/User       |
        +-------------------------+
          |   Login Data          \
          |   Control Commands     \
          |   Threshold Settings    \
          v                          \
    +-------------------------------------+
    |    Process 0: Smart Soil            |
    |        Irrigation System            |
    +-------------------------------------+
          ^                           /
          |  Soil Moisture Data      /
          |  Pump Status            /
          |  Alerts                /
          |  History              /
        +-------------------------+
        |       Farmer/User       |
        +-------------------------+
```




DFD Level 2 – Decide Irrigation Action
Moisture Data
Threshold & Settings
Mode Commands
P2.1:
Compare Moisture
P2.2:
Check Mode
P2.3:
Safety Rules
Pump Command
Alerts/Status
Logging Info


State-Transition Diagram – Smart Soil Irrigation System
Idle/
Monitoring
Manual Command
Manual OFF
Low Moisture
Auto ON
Moisture OK
Manual
Control
Irrigating
Network Lost
Cloud Restored
Maint Cmd
Done
Reset
Fault Detected
Maintenance
Offline
Auto Mode
Fault
State

5. Supporting Information

This section contains additional material that supports the development, understanding, and evaluation of the Smart Soil Irrigation System but is not part of the core functional requirements. It may include background research, hardware specifications, formulas, calibration methods, sensor characteristics, and detailed data descriptions.

Supporting information is typically provided in appendices to avoid overloading the main SRS content and to keep the requirement sections focused.

5.1 Purpose of Supporting Information

- *To document research that justifies design decisions (for example, choice of sensor type, threshold ranges, microcontroller selection).*
- *To provide detailed definitions of all data items used in the system (data dictionary).*
- *To record any formulas or algorithms used (e.g., moisture conversion formula, runtime calculation).*
- *To give extended descriptions that may be useful for future improvements or maintenance.*

5.2 Organization

The supporting information for this project is organized into two main appendices:

- ***Appendix A – Background Research on Smart Soil Irrigation Systems, Sensors, and IoT***
- ***Appendix B – Data Dictionary (detailed description of each data element used in the system)***

These appendices can be expanded as the project progresses.

Appendix A – Background Research on:

This appendix provides a summary of the theoretical and practical research that supports the Smart Soil Irrigation System.

A.1 Smart Irrigation and Water Management

- *Importance of precision irrigation in agriculture*
- *Problems with traditional manual irrigation (over-watering, under-watering, labor-intensive)*
- *Advantages of automated and sensor-based irrigation:*
    - *Water saving*
    - *Balanced soil moisture*
    - *Enhanced crop productivity*

A.2 Soil Moisture Sensors

- *Types of sensors:*
    - *Resistive soil moisture sensors*
    - *Capacitive soil moisture sensors*
- *Working principles:*
    - *How sensor output correlates with soil water content*
- *Calibration methods:*
    - *Matching sensor readings with actual soil condition*
- *Factors that affect sensor accuracy:*
    - *Soil type (clay/sand/loam)*
    - *Temperature, salinity, and installation depth*

A.3 Microcontrollers and Wireless Communication

- *Microcontroller platforms:*
    - *Arduino, ESP8266, ESP32 and their features*
- *Why ESP/Arduino platform is suitable:*
    - *Low power consumption*
    - *Easy programming (C/C++)*
    - *Built-in Wi-Fi (in ESP boards)*
- *Overview of Wi-Fi network use in farms and greenhouses*

A.4 IoT Platforms for Remote Monitoring

- *Concept of IoT (Internet of Things) in agriculture*
- *Cloud platforms / dashboards for:*
  - *Real-time data visualization*
  - *Remote control*
  - *Storing historical data*
- *MQTT/HTTP communication protocols used to send data from device to cloud*

A.5 Renewable Energy for Agricultural Automation

- *Use of battery and solar panels to power small irrigation controllers*
- *Benefits:*
  - *Independent operation in remote locations*
  - *Reduced electricity bills*
- *Basic design considerations for solar-powered systems (panel size, battery capacity, charge controller).*

Appendix B – Data Dictionary

The Data Dictionary defines and describes each significant data item used in the Smart Soil Irrigation System. It provides ***name****,* ***type****,* ***range****,* ***description****, and* ***where it is used****.*

B.1 Key Data Elements

Data Name	***Type***	***Range / Exam***	***Description***	***Used In***
moisture_percenta	*Integer / Float*	*0–100 (%)*	*Soil water content express as percentage*	*SensorReading, Logic*
raw_sensor_value	*Integer*	*0–1023 (ADC)*	*Raw ADC value read from sensor pin*	*Firmware*
threshold_value	*Integer*	*0–100 (%)*	*Minimum required moistur before irrigation starts*	*Settings, Logic*
pump_state	*Enum/Stri*	*"ON" / "OFF"*	*Current status of pump*	*UI, Logs*
mode	*Enum/Stri*	*"AUTO" /*	*System operating mode*	*Control Logic*

Data Name	***Type***	***Range / Exam***	***Description***	***Used In***
		"MANUAL"		
alert_status	*Boolean*	*True/False*	*Indicates if alert is active (e.g., low moisture)*	*UI Alerts*
timestamp	*DateTime*	*e.g., "2025-03-10:30:00"*	*Time at which reading/eve occurred*	*Logs, History*
user_id	*Integer*	*Positive*	*Unique ID of user*	*User Table*
username	*String*	*e.g., "farmer01*	*Login identifier*	*Authentication*
password_hash	*String*	*Encrypted strin*	*Hashed password stored securely*	*Authentication*
device_id	*String/Int*	*e.g., "DEV-001*	*Unique irrigation system ID*	*Mapping logs to device*
trigger_type	*Enum/Stri*	*"AUTO" / "MANUAL"*	*Reason why pump was activated*	*PumpEvent Table*
duration_minutes	*Float*	*≥ 0*	*Time for which pump remained ON*	*Analysis reports*
sensor_status	*Enum/Stri*	*"OK" / "FAULT*	*Health of sensor based on diagnostics*	*Fault handling*
wifi_signal_streng	*Integer*	*In dBm*	*Strength of Wi-Fi connecti*	*Diagnostics pane*

You can add more entries as you refine your implementation.